

Functional requirements	Non functional requirements
1. To shorten the url 2. To redirect to the original website 3. Custom url generation 4. Link Expiration	1. Less latency 2. Having a unique sort url 3. Availability 4. Read heavy system 5. Unpridectibility and security

Estimation
Daily Users - 10 Million 1 User - 3 short URL + 2 redirect link Total Daily short URLs - 30 Million Total Daily Redirect links - 20 Million Read Write Operations - 300 Million reads/day URL Expiry Period - 10 years Average long URL - 500 bytes Short URL length - 7 bytes Read:Write Ratio = 10:1 Write QPS = 700 qps Read QPS = 7000 qps Storage Estimation Daily Storage = 18 GB/day For 10 Years = 65.7 TB



